

Electricity (Public Safety) Regulations, 2018

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IT is hereby notified that the Minister of Energy and Power Development has, in terms of section 65 of the Electricity Act [*Chapter 13:19*], made the following regulations:—

Title

1. These regulations may be cited as the Electricity (Public Safety) Regulations, 2018.

Interpretation

2. In these regulations—

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- “artisan” or “electrician” means any person who undertakes electrical work and is a holder of an electrical qualification and is certified to undertake electrical work as defined in section 4;
- “dead” means zero voltage with respect to earth and disconnected from any live system;
- “designated responsible official” means a person appointed to oversee the construction, maintenance, and or operation of a piece of infrastructure;
- “earthed” means, effectively connected to the general mass of earth in such a way as to ensure at all times an immediate discharge of electrical energy without danger;
- “electrical risk” means—
- (a) the danger or hazard to a person or animal of death, shock or injury caused directly by electricity or originating from an electrical source; or
 - (b) the risk of damage or loss to the property caused directly by electricity or originating from an electricity source;
- “electrically safe” means that all persons or animals or property are free from electrical risk;
- “electricity source” means a point or a source on a gadget or property that emits electrical current;
- “electrical work” includes doing or supervising any construction, manufacturing, assembling, installing or connecting, maintaining, testing, certifying, or inspecting an electrical source;
- “inspector” means a person appointed as an inspector in terms of section 62 of the Act;
- “Zimbabwe Standard” or “ZWS” means a Zimbabwean standard approved by the Standards Association of Zimbabwe (SAZ).

Electrical works

3. (1) No designated responsible official or artisan shall do anything, or permit anything to be done, in the course of carrying out any electrical work, that results, or is likely to expose electrical risk to the public, employees or consumers.

(2) No person must perform or supervise electrical work unless—

- (a) he or she is a holder of a minimum Class IV Electrician Journeyman/Apprenticeship certificate issued by the Apprentice Board of Zimbabwe under the Manpower Planning and Development Act[Chapter 28:02] as prescribed in First Schedule; and
- (b) he or she has been trained on any potential or actual danger to health or safety in the workplace in relation to the work.

(3) No licensee or inspector shall approve electrical works or connection unless the electrical work has been carried out by—

- (a) a Class I certified Journeyman; and
- (b) the works have been done in terms of ZWS 400:2006 Electrical wiring of premises (ZWS wiring rules).

(4) All designated responsible officials or artisans must ensure that—

- (a) all work areas are electrically safe;
- (b) all electrical equipment and gadgets used in the conduct of the electrical work or undertaking are electrically safe;
- (c) during his or her conduct or undertaking, other persons who may be directly affected by such works or activities are not exposed to electrical risk;
- (d) he or she complies with—
 - (i) the electricity service providers instructions for the electrical safety of persons and property; and
 - (ii) ZWS 400:2006 Electrical wiring of premises (ZWS wiring rules); and
 - (iii) the Electricity (Wiring) Regulations, 1961;
 - (iv) approved standards, codes, engineering and safety standards or instructions governing the works;
- (e) he or she does not wilfully or negligently interfere with or misuse equipment put in place for electrical safety.

(5) Switching operations must be in accordance with the licensee's safety rules and subsequent amendments and artisans or

electricians must exercise extreme caution to avoid incorrect switching operations which give rise to electrical accidents.

(6) For the avoidance of doubt unless it is live line maintenance—

- (a) work shall be done on dead, isolated and earthed equipment; and
- (b) testing for dead shall be done before earthing.

(7) The designated official of electrical works being undertaken must ensure that all work is carried out safely in accordance with the licensee's safety rules.

(8) Any person who contravenes this section shall be guilty of an offence and liable to a fine not exceeding level 5 or, in default of payment, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

Prohibition of unauthorised electricity connections

4. (1) No electrical installation work, including, but not limited, to additions, alterations and repairs to existing installations, shall be carried out for the purpose of supply to such consumer except by an artisan as provided in section 3.

(2) No owner of property or consumers shall energise electrical installation work, including additions, alterations and repairs to existing installations, carried out on the premises of any consumer unless it has been approved by the licensee.

(3) The property owner must ensure that there are no illegal connections on their property at all times.

(4) Any person who contravenes this section shall be guilty of an offence and liable to a fine not exceeding level 10 or, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

Transmission and distribution infrastructure maintenance

5. (1) Any licensee or electricity undertaking whose infrastructure exposes the public to electrical risk shall—

- (a) ensure that it is electrically safe;

- (b) develop and implement asset design, procurement, operation and maintenance plans to eliminate the electrical risk;
- (c) carry out periodic inspections and maintenance of electrical infrastructure including testing of protection equipment in accordance with the plans referred to in paragraph (b);
- (d) keep records of such inspections, maintenance and inspections referred to in paragraphs (b) and (c) above for a period of five years;
- (e) rectify without delay any condition of infrastructure that exposes the public, animals or property to electrical risk on notification of such a condition;
- (f) maintain a public liability insurance policy to cover claims arising from such risk;
- (g) inform the victims of electrical accidents or their next of kin of their rights in terms of the policy referred to in paragraph (f);
- (h) ensure payout is made timeously on conviction or report.

(2) Any person who contravenes this section shall be guilty of an offence and liable to a fine not exceeding level 14 or, in default of payment, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

Licensee employee obligation

6. (1) No employee of a licensee or undertaking shall without reasonable cause or negligently do anything that is likely to endanger the electrical safety of the public or other employees.

(2) Whenever a manager, agent, contractor or employee of a licensee or undertaking does or omits to do anything which would constitute an offence for the licensee or undertaking, he or she shall be liable in his or her own right and if convicted and sentenced he or she must be convicted or sentenced in respect thereof as if he or she were the licensee or undertaking.

Periodic inspections of existing connections

7. (1) Where an installation is already connected to the supply system of the licensee, every such installation shall be periodically

inspected by licensee inspectors at intervals as prescribed in the Second Schedule.

(2) Where an inspector finds that an installation is no longer electrically safe, he or she shall immediately disconnect and isolate such an installation from the main supply until the electrical risk has been rectified.

(3) Where a person reconnects supply to an installation whose risk has not yet been rectified, such person shall be guilty of an offence and liable to a fine not exceeding level 5 or, in default of payment, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

(4) Where a licensee does not comply with subsections (1) and (2) such licensee shall be guilty of an offence and liable to a fine not exceeding level 14 or, in default of payment, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

Underground cables, conductors and equipment

8. (1) Every designated responsible official or artisan must ensure that the public is not endangered by underground cables, conductors and associated equipment during or after they undertake their electrical work by—

- (a) completely backfilling all cable trenches and the ground restored to its original state and the work site is cleared of any debris;
- (b) barricading works which are temporary in nature in respect of joints or terminations;
- (c) properly securing all outdoor substations, distribution boards and metering cubicles.

(2) Underground cables shall be kept at such depth as prescribed in the Third Schedule or be otherwise protected so as to avoid, so far as is reasonably practicable, any damage or danger by reason of such use of the land which can be reasonably expected.

(3) Before digging, boring, trenching, grading, excavating or breaking ground with tools, mechanical equipment or explosives, a contractor, owner or occupant of land, buildings or premises shall

ascertain from the licensee responsible for the distribution of electricity to the land, building or premises the location of any underground distribution cables that may be interfered within the course of such activities:

Provided that an appropriate cable avoidance tool must be used to detect any buried services.

(4) The licensee must keep up to date, a map or series of maps indicating the position and depth below surface level of all networks or parts thereof which he or she owns or operates and provide reasonable information with respect to the location of its underground distribution cables and associated plant at all times.

(5) The licensee must ensure all underground cables are laid in designated servitudes and where this is not possible the cable routes are clearly marked.

(6) The licensee shall carry out public awareness campaigns in all areas where electricity is distributed to make the public aware of the dangers of electricity power infrastructure and shall keep records of the campaigns.

(7) The licensee shall at all times have an emergency preparedness structure in place for reporting of dangerous electricity distribution infrastructure.

(8) Where a person contravenes the provisions of this section, such person shall be guilty of an offence and liable to a fine not exceeding level 12 or, in default of payment, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

Proximity to power lines (wayleaves)

9. (1) No person shall place an object, carry on any business or activity, farm or erect a structure within prescribed wayleave clearance distances to a conductor forming part of a system of overhead power lines as prescribed in the Fourth Schedule.

(2) A person may place an object or erect a structure closer to an energised conductor forming part of a system of overhead or underground distribution lines than the required minimum separations from energised conductors forming part of such a system if the person

first obtains a written authorisation from the distributor responsible for the energised conductor.

(3) Every licensee or undertaking, subject to subsection (5), must ensure that no trees or foliage grows to such an extent as to encroach and interfere with supply conductors.

(4) No person shall prohibit the licensee from complying with subsection (3).

(5) The licensee may exercise discretion with regard to clearing vegetation or infrastructure which would normally be outside the prescribed wayleave distance but is likely to interfere with power lines.

(6) Extreme caution must however be exercised during clearing operations, this includes de-energising the power line before undertaking clearing works.

(7) Where a person contravenes this section, such person shall be guilty of an offence and liable to a fine not exceeding level 5 or, in default of payment, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

Switched alternative sources of energy

10. (1) Where a person needs to operate a switched alternative source of energy not exceeding 100kW on a distribution licensee's network, he or she must ensure that that source of energy is operated with a change-over switch and the necessary protection required and the connection shall to be authorised by the distribution licensee to prevent back-feed into the distributor's network.

(2) Where a person operates a switched alternative source of energy without a change-over switch and without authority of the distribution licensee, such person shall be guilty of an offence and liable to a fine not exceeding level 5 or, in default of payment, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

Irrigation pipes and over-height loading

11. (1) Every employer or farm owner must induct employees on hazards associated with electricity, safe working procedures and precautions within the environment in which they work.

(2) No employer or farm owner shall permit the carrying of irrigation pipes in the upright position so as to encroach or make contact with power lines within his or her premises.

(3) The maximum loading height for all vehicular mode of transport by road shall be 5.5 metres:

Provided that abnormal load movement of height exceeding 5.5 metres shall seek clearance and escort from the local electricity supplier.

(4) Any person who contravenes this section shall be guilty of an offence and liable to a fine not exceeding level 10 or, in default of payment, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

Electric fences

12. (1) All owners of property and electrical contractors must ensure that electric fences are connected as specified in ZWS 859:2003 safety requirements for electric fence energisers.

(2) Where a person connects an electric in contravention of the section, such a person shall be guilty of an offence and liable to a fine not exceeding level 5 or, in default of payment, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

Security of electrical facilities

13. (1) Every employer or licensee must for every substation and reticulation system which they own or operate ensure—

- (a) all electricity facilities of works and installations, and all substations, are secured against access by unauthorised persons through a minimum height for fences and walls of 2.4m; and
- (b) all installations including mini substations shall be secured through lockable doors or metallic barricades; and
- (c) all pylons and poles carrying conductors have anti-climb devices where applicable as prescribed in the licensee's standards; and

- (d) that at all times at the entrance and around the substation, the following is be displayed—
 - (i) a danger warning sign as prescribed in the Electricity (Supply) Regulations, 1961;
 - (ii) a notice specifying the location or identification of the substation, the name of the licensee who owns or operates the equipment making up the substation and the telephone number where a suitably qualified person appointed for this purpose by the licensee will be in constant attendance;

(2) It shall be an offence to affix any material, including advertisement material on any electrical infrastructure.

(3) Where a licensee contravenes the provisions of this section, it shall be guilty of an offence and liable to a fine not exceeding level 5 or, in default of payment, imprisonment for a period not exceeding six months or both such fine and such imprisonment.

Cradle guards

14. (1) Every licensee through the responsible local official shall for every reticulation system which they own or operate ensure—

- (a) cradle guards are in place where electrical conductors cross telephone lines;
- (b) that there is enough clearance between conductors where conductors of different voltage levels cross or are in close proximity such as to prevent mutual inductance between the conductors.

FIRST SCHEDULE (Section 3(1))

Proficiency Levels for Electrical Apprenticeship Classes:

CLASS	PROFIECIENCY LEVEL	TRAINING DURATION (YEARS)
I	Fully qualified electrical artisan able to execute duties independently and supervise. Competent in duties covering all classes from I to IV.	4
II	Electrical artisan who can perform duties as instructed but not able to work independently-needs supervision. Can also work independently up to competencies involved. Proficient in duties in classes III and IV as well.	3
III	Electrical artisan who can perform basic duties under supervision including class IV duties	2
IV	Entrant level for electrical artisan who only performs duties under supervision	1

SECOND SCHEDULE (Section 7(1))

INSTALLATION INSPECTION FREQUENCY

Type of Installation	Maximum Period Between Inspections
(a) General	
<i>Domestic</i>	<i>10 years</i>
<i>Commercial, educational, hospitals</i>	<i>5 years</i>
<i>Industrial</i>	<i>3 years</i>
(b) Buildings open to the public	
<i>Cinemas, places of worship, leisure complexes</i>	<i>1 year</i>
<i>Places of public entertainment, restaurants, hotels, theatres</i>	<i>1 year</i>
(c) External installations	
<i>Agricultural and horticultural</i>	<i>3 years</i>
<i>Caravan sites</i>	<i>1 year</i>
<i>Highway power supplies</i>	<i>6 years</i>
(d) Special installations	
<i>Emergency lighting</i>	<i>3 years</i>
<i>Fire alarms</i>	<i>1 year</i>

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Type of Installation	Maximum Period Between Inspections
<i>Launderettes</i>	<i>1 year</i>
<i>Petrol/LPG filling stations</i>	<i>1 year</i>
<i>Temporary Installations</i>	<i>3 months</i>

THIRD SCHEDULE (Section 8(2))

CABLE BURIAL DEPTHS

	LV (mm)	11kV(mm)	33kV(mm)	132kV(mm)
Footways, grass verges or private property	450	600	900	900
Tarred roads (including road crossings)	600	750	900	900
Normal agricultural land (not subject to deep ploughing)	1,050	1,050	1,050	1,050
Agricultural land subject to deep ploughing	1,200	1,200	1,200	1,200

FOURTH SCHEDULE (Section 9(1))

WAYLEAVE CLEARANCES

Voltage Level	Wayleave Clearance (m) either side from the outermost conductor
11kV	5
22kV	5
33kV	7.5
66kV	15
88kV	15
132kV	15
330kV	30
400kV	30

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